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RESEARCH ARTICLE

Forecasting Provincial Healthcare Infrastructure and Budget Allocation in Turkey Using Machine Learning and TOPSIS

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Abstract
Equitable allocation of healthcare resources across provinces requires accurate projections of future infrastructure needs, demographic pressures, and available budgets. This study proposes an integrated machine learning framework for province-level healthcare planning in Turkey covering the 2025–2030 period. The framework consists of three interconnected components. In the first component, six healthcare infrastructure indicators the number of "physicians, nurses, midwives, other healthcare personnel, hospital beds, and hospitals" are forecasted for all 81 Turkish provinces using 19 machine learning models trained on 2002–2024 data with a five-fold walk-forward cross-validation scheme. In the second component, four demographic and service capacity criteria "the 0–14 age group ratio, the 65+ age group ratio, the supply of physicians per 1,000 people, and the supply of hospital beds per 100,000 people" are projected to 2030 and used to construct TOPSIS-based provincial need rankings. Criterion weights are derived objectively using the Shannon entropy method. In the third component, Turkey's national health budget is forecasted in USD terms using an ensemble of time-series and regression models, and the results are converted to Turkish lira using projected exchange rates. The three components are integrated through a dynamic budget allocation mechanism with a feedback loop: after each annual allocation, provincial TOPSIS scores are updated to reflect received resources, ensuring that high-priority provinces in early years yield priority to underserved provinces in subsequent years. Results show that small northern provinces with high elderly population ratios and low physician density, such as Sinop and Artvin, rank highest in 2025, while the feedback mechanism redistributes priority to larger coastal provinces from 2026 onwards. The proposed framework provides a data-driven, adaptive, and reproducible approach to provincial health resource planning.

Keywords: Budget Allocation, Healthcare Infrastructure Forecasting, Machine Learning, Shannon Entropy, TOPSIS

1. Introduction

Province-level healthcare resource planning in Turkey faces a structural challenge. Personnel assignments are still largely based on previous vacant position records, the departure of personnel who have completed their service period, or rotations initiated by provincial health directorates [1]. Although this approach may address the personnel shortage of a particular hospital, it may also lead to workforce deficits in hospitals that have a greater need for personnel but are less preferred by healthcare workers. Budget management in healthcare is a fundamental component of health planning. One of the major challenges in Turkey's healthcare planning process is budget allocation. In general, a larger share of healthcare resources is allocated to metropolitan areas due to their higher population density. However, this situation may lead to financial constraints in smaller cities that also require adequate healthcare funding, thereby weakening residents' access to healthcare services [2]. Turkey provides broad access to publicly financed healthcare [3], which places a substantial planning responsibility on the Ministry of Health, requiring forward-looking, data-driven tools rather than reactive workforce management.

The issue of healthcare planning in Turkey is addressed not only through the initiatives of the Ministry of Health but also through academic and independent studies. Various studies have been conducted on this topic, offering insights into how specific aspects of healthcare planning can be improved. Among these studies, some evaluate the distribution of healthcare personnel in Turkey, while others examine the economic dimension of healthcare planning [4, 5].

Healthcare planning has gained a new perspective with the emergence of artificial intelligence (AI). In particular, AI-based forecasting and allocation approaches have become increasingly prominent in recent

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planning studies. Using AI-based methods, cities with greater personnel needs can be identified more effectively, and the criteria guiding budget allocation can be determined in a more systematic, data-driven manner.

In recent years, AI-based personnel allocation forecasting has become a prominent research topic in the literature. In one study, the physician supply and demand balance in Turkey for 2030 was modeled using XGBoost, yielding the most robust forecasts [6]. In another, nurse staffing adequacy and patient care models in hospital wards were evaluated using machine learning, with Random Forest achieving the highest performance [7]. However, both studies address workforce forecasting in isolation and do not link their projections to a need-based provincial ranking or a budget distribution mechanism.

In personnel forecasting, population balance is a key factor; however, demographic structure must also be evaluated independently of population size. The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method offers a suitable framework for examining such regional and demographic differences. A TOPSIS-based study evaluating the sustainable development performance of TR33 region provinces across housing, transportation, urbanization, and environmental criteria demonstrated that multidimensional city-level ranking can be effectively applied in healthcare personnel planning as well [8]. Entropy-weighted TOPSIS has become an established approach in health system evaluation. It has been used to assess the equity of regional health resource allocation [9], [10], to rank the service quality of tertiary public hospitals at city level [11], and to evaluate community-level basic public health services [12]. At the cross-country level, the same method has been applied to integrate equity measures across demographic, geographic, and economic dimensions in ASEAN health systems [13], while in the Turkish context it was used to compare the performance of 544 secondary-level public hospitals across geographical regions [14]. These studies confirm that entropy-based objective weighting is well suited to health-sector indicators, where subjective expert weights are difficult to justify.

Such evaluations, however, describe allocation as it currently stands rather than projecting future need, and spatial inequalities in healthcare resources have been shown to persist over time when allocation rules remain fixed [15]. Nevertheless, such ranking studies remain static and do not incorporate forecasted infrastructure indicators or a dynamic resource feedback mechanism across planning periods.

In healthcare planning, studies have used AI for budget planning. In one study, Turkey's per capita healthcare expenditures were estimated using AI techniques and genetic algorithm-based feature selection, yielding the best-performing results with the Random Forest model [16]. In another study focusing on per capita healthcare expenditure, Turkey's healthcare expenditures were predicted using machine learning models, with the Support Vector Regression algorithm achieving the highest performance [17]. Yet both studies operate at the national aggregate level and provide no mechanism for translating budget forecasts into province-level allocations.

This study extends our previous research, in which the regional distribution of healthcare workforce demand in Turkey was forecasted using ML algorithms based on 2024 personnel distribution patterns [18]. However, the previous study did not incorporate demographic population characteristics, TOPSIS-based need prioritization, healthcare budget allocation, or improved machine learning methods. Therefore, the present study aims to provide a more comprehensive healthcare planning framework by integrating personnel distribution forecasts, demographic indicators, TOPSIS scores, and machine learning-based healthcare budget allocation for the 2025–2030 period. Table 1 summarises the methodological scope of the studies discussed above and positions the present work with respect to them.

The main contributions of this study are as follows:

- A province-level infrastructure forecast for all 81 Turkish provinces across six indicators using 19 machine learning models with walk-forward cross-validation.
- An entropy-weighted dynamic TOPSIS need ranking updated annually via a feedback loop that prevents budget monopolisation by early-priority provinces.
- A national health budget forecast in USD integrated with province-level allocations for the 2025–2030 period.

Table 1: Comparison of Related Studies and the Positioning of the Present Work

Study	Method and Scope	Missing Component
[6]	XGBoost, national	No need ranking or allocation
[7]	Random Forest, hospital ward	No spatial dimension
[16]	RF with genetic algorithm, national	Expenditure only
[17]	SVR, GPR, DT, national	Expenditure only
[8]	TOPSIS, regional	Static, observed data only
[14]	Entropy–TOPSIS, national	Static, retrospective
[13]	Entropy–TOPSIS with DEA, cross-country	Static, no forecasting
[9], [10]	Entropy–TOPSIS, regional	Static, no budget link
[18]	ML forecasting, province	No demographics, MCDM or budget
This study	19 ML models, entropy–TOPSIS, dynamic feedback, province	—

2. Material and Methods

In the relevant section of the study, the data sources used for the dataset, the preprocessing steps applied to the datasets, the pre-modeling preparation stages, the preferred machine learning (ML) approaches, weighted ensemble selection strategies, and the model evaluation metrics are explained in detail. The explanations are presented under separate headings to improve clarity.

The overall workflow of the proposed framework, together with the data flow between its three modules, is presented in Figure 1.

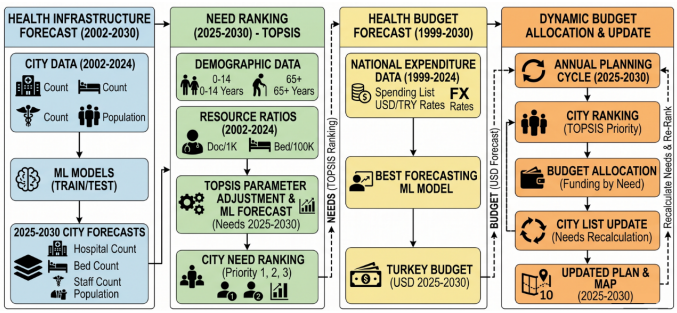


Figure 1: Overall workflow of the proposed healthcare planning framework, showing the infrastructure forecasting, need prioritization, and budget allocation modules and the dynamic feedback loop connecting them

2.1. Data Collection Sources

This section describes the datasets used in the study. It explains the sources from which the datasets were collected and the types of data obtained from these sources. The subsequent analysis and processing steps applied to the collected data are described in detail in other sections.

2.1.1. Dataset Sources

The data used in this study were collected from three main source groups (Figure 2). The primary source was the Turkish Statistical Institute (TURKSTAT), which provided demographic and healthcare-

related statistics [19]. The second source was Investing.com, from which historical USD/TRY exchange rate data were obtained. The third source group was used to obtain actual and forecasted USD/TRY exchange rate data for future periods.

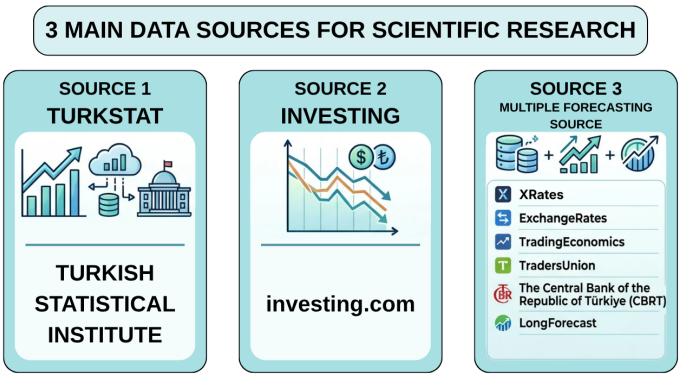


Figure 2: The Three Main Source Groups from Which the Datasets Were Collected

2.1.2. Data Collected from the Data Sources

After reliable data sources were identified in Section 2.1.1, the required datasets were collected in accordance with the study's intent. The collected data are summarized (Table 2 and Table 3). The first table presents the source, spatial coverage, and year range of each dataset. In addition, the second table provides detailed information on the content and size of the corresponding data. The corresponding module associated with each dataset is also included in the tables. The purpose of each dataset and its role within the overall framework of the study are explained under Section 3 in the Results section.

Table 2: Data Sources and Spatial Coverage of Collected Datasets

Module	Data Source	Spatial Level
Infrastructure Forecasting	TURKSTAT	Province
Need Prioritization	TURKSTAT	Province
Need Prioritization	TURKSTAT	Province
Need Prioritization	TURKSTAT	Province
Budget Allocation	TURKSTAT	Country
Budget Allocation	Investing.com	Country
Budget Allocation	Third Group	Country

Table 2 shows that the datasets are drawn from two primary sources: TURKSTAT and external financial data providers. Province-level data cover the 81 provinces of Turkey and are used in the Infrastructure Forecasting Module (IFM) and Need Prioritization modules, while country-level data support the macroeconomic analyses conducted in the Budget Allocation module. This division reflects the spatial resolution at which each quantity is defined, since infrastructure and demographic indicators vary across provinces while expenditure and exchange rate data are reported only at the national level.

Table 3 presents the temporal scope and content of each dataset. The largest dataset comprises 13,041 data points, corresponding to 1,863 province-year observations across seven variables for the 2002–2024 period, while the smallest comprises 25 annual country-level health expenditure records.

It should be noted that some datasets contain multiple sub-variables within a single data type. In particular, the healthcare personnel dataset comprises six distinct personnel categories: assistant physicians, specialist physicians, general practitioners, midwives, nurses, and other healthcare staff.

Table 3: Year Range, Data Types, and Dataset Sizes

Year Range	Data Types	Dataset Size
2002–2024	Population, number of hospitals, number of hospital beds, and healthcare personnel data	13041
2025, 2030	Population data for 2025 and projected population data for 2030	162
2007–2025	Population data for the 0–14 and 65+ age groups	3078
2002–2024	Supply of physicians per thousand people and supply of hospital beds per one hundred thousand people	3726
1999–2024	General health expenditure data	25
1999–2024	Monthly USD/TRY exchange rate data	300
2025–2030	Actual and forecasted USD/TRY exchange rate data	72

2.2. Data Preprocessing

Data preprocessing is the process of cleaning, transforming, and integrating raw data before analysis so that it can be processed more effectively and accurately by data mining algorithms. This process includes fundamental steps such as handling missing or noisy data through data cleaning, converting data into appropriate formats through data transformation, and removing unnecessary features from the dataset through data reduction. In this context, data preprocessing aims to maximize data quality and directly improve the accuracy, reliability, and performance of the ML or data mining models to be developed [20].

In this study, the data preprocessing stage was carried out to enable machine learning algorithms to operate more effectively on the data. All procedures performed during the data preprocessing stage were organized into subsections according to the sources from which the data were obtained. Since no additional preprocessing was required for the Investing.com dataset group, preprocessing procedures were applied only to the TURKSTAT datasets and the third dataset group.

2.2.1. TURKSTAT Dataset Preprocessing

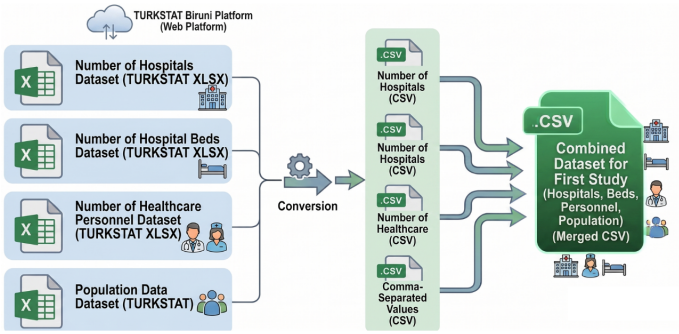


Figure 3: Merging of the Four TURKSTAT Datasets Used in the IFM

TURKSTAT data were collected separately in Office Open XML Spreadsheet (XLSX) format from the web platform named Biruni, which provides access to TURKSTAT datasets [21]. The collected datasets were then converted into Comma-Separated Values (CSV) format, which can be processed more conveniently by machine learning models [22]. After the conversion process, the datasets were merged into a common file according to the study group in which they were used. For example, since the IFM used data on the number of hospitals, hospital beds, healthcare personnel, and population, these four datasets were combined into a single file (Figure 3).

2.2.2. Third Group Dataset Preprocessing

The third dataset group consisted of sources that provide USD/TRY exchange rate data as well as future exchange rate forecasts. The data in this group were directly collected in CSV format using a Python-based web scraper [23]. Subsequently, similar to the preprocessing procedure applied to the TURKSTAT datasets, the collected data were merged into a single CSV file in a suitable format to improve their usability for ML models [22]. Figure 4 illustrates the data extraction process using the web scraper and the integration of the collected data into a single CSV file.

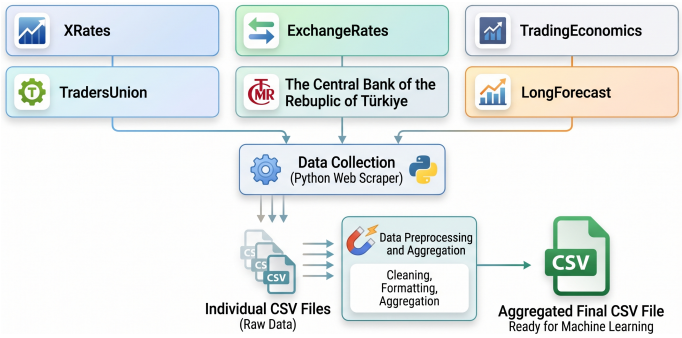


Figure 4: Extraction and Aggregation of USD/TRY Exchange Rate Data Using a Python-Based Web Scraper

2.3. Pre-Modeling Preparation

Before proceeding to the prediction stage, several preparatory steps are required for ML models. These preparation stages are generally referred to as pre-modeling [24]. In the pre-modeling stage, six different preparatory steps were performed. These steps are illustrated in Figure 5 and then explained in detail under separate subsections.

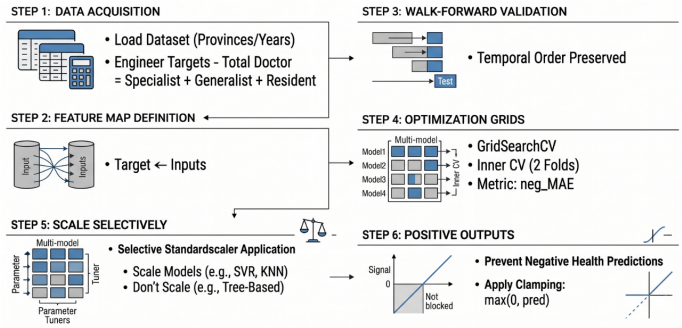


Figure 5: The Six Pre-Modeling Preparation Steps Applied Before Model Training

2.3.1. Data Loading and Variable Derivation

The dataset used in the IFM was compiled from province-level health records spanning the years 2002 to 2024, covering all 81 provinces of Turkey. The resulting dataset consists of $81 \times 23 = 1{,}863$ observations, which can be expressed as:

$$N = P \times T = 81 \times 23 = 1{,}863 \quad (1)$$

where N denotes the total number of observations, P is the number of provinces, and T is the number of years in the study period. In addition, a composite variable representing the total number of physicians was derived by aggregating three sub-categories of healthcare personnel available in the dataset. The derived variable is defined in Equation 2 as follows:

$$\text{Physician} = \text{Specialist} + \text{General Practitioner} + \text{Resident} \quad (2)$$

This aggregation was preferred over using each sub-category separately, as the distribution of residents across provinces exhibits considerable instability and abrupt fluctuations over time. The composite variable therefore provides a more stable and consistent representation of total physician availability at the province level.

2.3.2. Feature Map

Feature extraction is a critical process in the use of ML, as the selection of input variables directly influences model performance [25]. In this study, a target-specific feature map was constructed, where each target variable is predicted using all remaining health infrastructure variables along with year and population as inputs. In other words, each target variable serves as an output while all other variables act as its predictors, ensuring that no target variable is used to predict itself. To illustrate the corresponding feature map structure applied in the study, the obtained feature assignments are presented in Figure 6.

```
feature_map = {
    "doctor":   ["year", "population", "hospital", "bed", "nurse", "midwife", "other"],
    "midwife":  ["year", "population", "hospital", "bed", "doctor", "nurse", "other"],
    "other":    ["year", "population", "hospital", "bed", "doctor", "nurse", "midwife"],
    "nurse":    ["year", "population", "hospital", "bed", "doctor", "midwife", "other"],
    "bed":      ["year", "population", "hospital", "doctor", "nurse", "midwife", "other"],
    "hospital": ["year", "population", "bed", "doctor", "nurse", "midwife", "other"],
}
```

Figure 6: Target-Specific Feature Map: Input Variables Assigned to Each Target

2.3.3. Walk-Forward Cross-Validation

Cross-validation is a resampling technique that estimates model generalization performance by partitioning data into k subsets, using each as a test set once while training on the remainder [26]. Empirical studies suggest that $k = 5$ or $k = 10$ generally yields reliable estimates [27]. However, standard random k-fold cross-validation is not suitable for time-series data, as randomly assigning observations to folds can cause future data to leak into the training set. To address this, a walk-forward strategy with an expanding training window was adopted, preserving the temporal order of observations across five folds as shown in Table 4.

Table 4: Training and Test Periods of the Five Walk-Forward Folds

Fold	Training Period	Test Period
1	2002–2016	2017–2020
2	2002–2017	2018–2021
3	2002–2018	2019–2022
4	2002–2019	2020–2023
5	2002–2020	2021–2024

Fold 5 serves as the primary evaluation fold, as its test period covers the years closest to the forecast horizon of the study.

2.3.4. Hyperparameter Optimization

Hyperparameters are configuration settings that govern the learning process of a model and must be specified prior to training, as they cannot be learned directly from the data [28]. The goal of hyperparameter optimization is to find the configuration λ^* that minimizes a predefined loss function over a search space Λ:

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \mathscr{L}\left(f_\lambda\left(X_{\text{train}}\right), X_{\text{val}}\right) \tag{3}$$

where f_λ denotes the model trained with hyperparameter configuration λ, and $\mathscr{L}$ is the loss function evaluated on the validation set.
In this study, Grid Search Cross-Validation (GridSearchCV) was employed, which exhaustively evaluates all possible combinations within a predefined hyperparameter grid [29]. For a grid containing m hyperparameters with n_i candidate values for the i-th hyperparameter, the total number of configurations evaluated is:

$$|\Lambda| = \prod_{i=1}^{m} n_i \tag{4}$$

Each configuration $\lambda \in \Lambda$ is evaluated using k-fold cross-validation with $k = 2$ inner folds, and the mean validation loss across folds is computed as:

$$\overline{\mathscr{L}}(\lambda) = \frac{1}{k} \sum_{j=1}^{k} \mathscr{L}\left(f_\lambda^{(j)}\right) \tag{5}$$

where $f_\lambda^{(j)}$ denotes the model trained on fold j. The scoring criterion used was the negative mean absolute error ($-$MAE), so the optimal configuration satisfies:

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \overline{\mathscr{L}}(\lambda) = \arg\min_{\lambda \in \Lambda} \frac{1}{k} \sum_{j=1}^{k} \frac{1}{n_j} \sum_{i=1}^{n_j} \left| y_i - \hat{y}_i^{(\lambda, j)} \right| \tag{6}$$

where n_j is the number of observations in fold j, y_i is the true value, and $\hat{y}_i^{(\lambda,j)}$ is the predicted value under configuration λ in fold j.
The search grids and the resulting selections are summarised in Table 5.

Table 5: Hyperparameter Search Grids and Values Selected at Fold 5

Model	Search grid	Selected (Fold 5)
LR	fit_intercept: {True, False}	Target-specific
RF	n_estimators: {100, 200, 300}	300
	max_depth: {None, 10}	10
ExtraTrees	n_estimators: {100, 200, 300}	Target-specific
	max_depth: {None, 10}	10
GB	n_estimators: {100, 200}	200
	learning_rate: {0.05, 0.1}	0.05
	max_depth: {3, 5}	3
XGBoost	n_estimators: {200, 300}	200
	learning_rate: {0.05, 0.1}	0.05
	max_depth: {3, 6}	3
CatBoost	iterations: {200, 300}	300
	depth: {4, 6, 8}	8
Ridge	Not tuned[a]	Default
ElasticNet	Not tuned[a]	Default
HGB	Not tuned[a]	Default

Selected values are the most frequently chosen setting across the six target variables; "target-specific" indicates no single setting was dominant.
[a] Applied with the default settings of the corresponding library implementation.

Grid search was applied with two inner folds, negative mean absolute error as the scoring criterion, and a fixed random seed of 42 to ensure reproducibility. Because optimisation was performed independently for each target variable and each outer fold, the reported selections correspond to the setting most frequently chosen across the six targets at Fold 5. Time-series and baseline models were excluded from grid search, as their parameters are estimated directly from the series rather than selected from a predefined grid.

2.3.5. Feature Scaling

Feature scaling is a preprocessing step that brings all input variables to a common numerical range, which is particularly critical for regularized linear models such as Ridge, as the penalty terms of these models are sensitive to the magnitude input features [30]. In this study, StandardScaler was applied exclusively to LR, and Ridge, while tree-based and boosting models were left unscaled as they are invariant to monotonic feature transformations. StandardScaler standardizes each feature by removing the mean and scaling to unit variance:

$$z = \frac{x - \mu}{\sigma} \tag{7}$$

where x denotes original feature value, while μ and σ represent the mean and standard deviation of training set. The scaler was fitted only on the training fold and then applied to test fold to avoid data leakage.

2.3.6. Non-Negativity Constraint

Since all target variables in this study represent counts of physical or human health resources, such as the number of hospitals, beds, or personnel, negative predicted values are inherently meaningless. To ensure the physical validity of all forecasts, a non-negativity constraint was applied by clipping all model outputs at zero, such that any predicted value $\hat{y} < 0$ was replaced by zero:

$$\hat{y} = \max(0, \hat{y}) \tag{8}$$

This constraint was applied uniformly across all models and target variables following each prediction step.

2.4. Machine Learning Approaches

ML is the science of algorithms that enable computer systems to improve their task performance by learning from data-driven experience rather than being explicitly programmed. This discipline focuses on constructing statistical models capable of discovering structural patterns hidden within large datasets and making accurate predictions about future uncertainties or unseen cases. The main objective of this process is to enable a system to successfully analyze new examples that it has not previously encountered by applying the rules and generalizations learned from historical data [31].

Table 6: Machine Learning Model Groups and Their Assigned Modules

Module	Main Category	Models Included
M1, M2	Classical Regression Models	LinearRegression (LR), Ridge, ElasticNet
M1, M2, M3	Tree-Based Models	RandomForest (RF), ExtraTrees, GradientBoosting (GB), HistGradientBoosting (HGB), XGBoost, CatBoost
M1, M2, M3	Time-Series Models	ARIMA, Prophet, HoltWinters
M3	Time-Series Models	Holt, HoltDamped
M1, M2	Ensemble Models	Voting, Weighted Ensemble
M2, M3	Baseline/Trend Models	Baseline_Lag1, CAGR5, CAGR10

M1: Infrastructure Forecasting Module (IFM), M2: Need Prioritization, M3: Budget Allocation. These labels and the module names are used interchangeably.

In this study, a total of nineteen different ML algorithms were used across three modules. These algorithms were employed to compare the prediction results and thereby identify the approach that produced the highest prediction performance. The ML algorithms used in the study are presented in Table 6 according to their names and the corresponding module.

2.4.1. Classical Regression Models

Classical regression methods are statistical techniques used to model the linear association between a response variable and one or more explanatory variables by estimating a parametric function from the available data. These models assume that the relationship between variables can be adequately described by a linear combination of predictors and rely on well-established estimation techniques. Despite their relative simplicity, they offer high interpretability and serve as important benchmarks against which more complex methods can be evaluated. Three classical regression models are employed in this study.

- Linear Regression estimates the dependent variable as a linear combination of independent variables using ordinary least squares (OLS), which minimizes the sum of squared residuals [32].
- Ridge Regression extends OLS by adding an L2 penalty term to reduce multicollinearity. It shrinks coefficients toward zero without setting them exactly to zero [33].
- ElasticNet combines L1 and L2 penalty terms, offering both variable selection and multicollinearity control, merging the advantages of Ridge and LASSO [33].

The advantages and disadvantages of the classical regression models are presented in Table 7 .

Table 7: Advantages and Disadvantages of Classical Regression Models

Model	Advantages	Disadvantages
Linear Regression	Simple to interpret; low computational cost	Sensitive to multicollinearity and outliers
Ridge	Controls multicollinearity; retains all variables	No variable selection; coefficients never reach zero
ElasticNet	Performs variable selection and controls multicollinearity	Requires tuning of two hyperparameters

2.4.2. Tree-Based Models

Tree-based models are non-parametric ensemble methods that partition the feature space through a series of recursive binary splits. Each split is determined by selecting the feature and threshold that best separates the target variable, allowing the model to approximate complex, non-linear decision boundaries without requiring feature scaling or distributional assumptions. These models have been shown to consistently outperform deep learning approaches on typical tabular datasets, owing to their robustness to irrelevant features, their ability to handle mixed data types, and their capacity to capture interaction effects without extensive preprocessing [34].

- RF constructs multiple decision trees using bootstrap sampling and random feature subsets, then combines their predictions to reduce variance and improve generalization ability. [34].
- ExtraTrees extends the random forest idea by selecting split thresholds randomly rather than optimally, which further reduces variance at the cost of a slight increase in bias [35].
- GB generates trees sequentially; each tree corrects the remaining errors of the previous tree by following the negative slope of the loss function. [34].
- HGB is a histogram-based variant of GB that discretizes continuous features into bins before splitting, substantially reducing training time on large datasets [34].
- XGBoost is a regularized and computationally optimized GB framework that supports parallel processing and natively handles missing values [34].
- CatBoost is a GB algorithm specifically designed to handle categorical features without manual encoding, using ordered boosting to prevent target leakage [36].

The advantages and disadvantages of the tree-based models are presented in Table 8 .

Table 8: Advantages and Disadvantages of Tree-Based Models

Model	Advantages	Disadvantages
RF	Low variance; robust to noise and outliers	High memory usage; slow on large datasets
ExtraTrees	Faster training than RF; further variance reduction	Slightly higher bias; less interpretable
GB	High predictive accuracy; handles mixed data types	Prone to overfitting; slow sequential training
HGB	Fast on large datasets; native missing value support	Less interpretable; requires bin size tuning
XGBoost	Regularization reduces overfitting; parallelizable	Many hyperparameters; sensitive to tuning
CatBoost	Native categorical support; robust to overfitting	Slower training than XGBoost; high memory usage

2.4.3. Time-Series Models

Time-series models are developed to represent temporal patterns and trend structures in ordered sequential data. Unlike cross-sectional methods, these models explicitly account for the order of observations and the patterns that evolve over time. Five time-series models are employed in this study.

- ARIMA (Autoregressive Integrated Moving Average) models a time series through autoregressive and moving average components after differencing to achieve stationarity [37].
- Prophet is a decomposable forecasting model developed by Meta that separately models trend, seasonality, and holiday effects, making it robust to missing data and structural breaks [37].
- Holt-Winters extends simple exponential smoothing by incorporating both trend and seasonal components through double or triple smoothing equations [38].
- Holt applies double exponential smoothing to capture level and trend in non-seasonal time series, using two smoothing parameters [38].
- Holt Damped introduces a damping parameter to the Holt method, moderating the trend component over the forecast horizon to avoid over-optimistic long-range predictions [38].

Table 9: Advantages and Disadvantages of Time-Series Models

Model	Advantages	Disadvantages
ARIMA	Flexible for stationary series; well-established statistical foundation	Requires stationarity; struggles with non-linear patterns
Prophet	Handles missing data and outliers; easy to configure	Less effective on short or irregular series
Holt-Winters	Captures both trend and seasonality; interpretable parameters	Assumes fixed seasonal pattern; sensitive to outliers
Holt	Simple and effective for trended non-seasonal data	Cannot capture seasonality; may overfit short series
Holt Damped	Reduces over-forecasting in long horizons; more conservative	Requires tuning of the damping parameter

The advantages and disadvantages of the time-series models are presented in Table 9 .

2.4.4. Ensemble Models

Ensemble models integrate the outputs of several base learners to improve generalization compared with an individual model. By combining models with different characteristics, ensemble techniques can lower variance, bias, or both [39]. In this study, two ensemble approaches are used.

- Voting Ensemble aggregates the predictions of multiple independently trained models by averaging their outputs. In this study, RF, XGBoost, and GB are combined using soft voting [39].
- Weighted Ensemble assigns optimized weights to each base model's prediction rather than treating all models equally. Weights are determined by minimizing validation error, allowing better-performing models to contribute more to the final prediction [40].

The advantages and disadvantages of the ensemble models are presented in Table 10 .

Table 10: Advantages and Disadvantages of Ensemble Models

Model	Advantages	Disadvantages
Voting	Simple to implement; reduces individual model errors	Equal weighting may dilute strong models
Weighted Ensemble	Optimized contribution of each model; adaptive	Weight optimization requires careful validation strategy

2.4.5. Baseline and Trend Models

Baseline and trend models provide simple yet informative reference points against which the performance of more complex models can be assessed. Rather than learning intricate patterns from data, these models rely on straightforward statistical rules or growth rate calculations. They are particularly useful for evaluating whether sophisticated algorithms offer meaningful improvements over naive forecasting strategies. Three baseline and trend models are employed in this study.

- Baseline Lag1 is a naive forecasting model that uses the most recent observed value as the prediction for the next period. Despite its simplicity, it serves as a fundamental benchmark in time-series forecasting tasks [41].
- **CAGR5** estimates future values by applying the compound annual growth rate calculated over the preceding five years. This approach captures medium-term growth momentum and provides a trend-based projection without model fitting [42].
- **CAGR10** follows the same logic as CAGR5 but extends the calculation window to ten years, offering a longer-term perspective on growth trends and reducing the influence of short-term fluctuations [42].

Table 11: Advantages and Disadvantages of Baseline and Trend Models

Model	Advantages	Disadvantages
Baseline Lag1	Simple to implement; requires no training	Ignores long-term trends; poor on volatile series
CAGR5	Captures medium-term growth; interpretable	Sensitive to the chosen base year; assumes constant growth
CAGR10	Smooths short-term fluctuations; stable long-term estimate	May miss recent structural changes; assumes constant growth

The advantages and disadvantages of the baseline and trend models are presented in Table 11 .

2.5. Weighted Ensemble Selection Strategy

While Section 2.4.4 introduced the general ensemble modeling approaches employed in this study, the present section describes the specific ensemble selection strategy applied during the model evaluation phase. Rather than using a fixed ensemble configuration, a data-driven weighted ensemble was constructed separately for each target variable based on cross-validation performance.

Ensemble methods combine the predictions of multiple base learners to achieve lower generalization error than any individual model alone [43]. The underlying principle is that diverse models tend to make different errors, and their aggregation reduces the overall prediction variance [44]. Ensemble approaches have been shown to consistently outperform single models across a wide range of regression and classification tasks [45].
In this study, for each target variable y_k, the top-K models were selected based on their Fold-5 accuracy scores, and their predictions were combined through a normalized weighted average. Let $\mathscr{M}_k = \{m_1, m_2, \ldots, m_K\}$ denote the set of top-K models selected for target y_k. The weight assigned to model m_i is given in Equation 9:

$$w_i = \frac{\text{Accuracy}_i}{\sum_{j=1}^{K} \text{Accuracy}_j}, \quad \sum_{i=1}^{K} w_i = 1 \tag{9}$$

The ensemble prediction for target y_k is then computed as:

$$\hat{y}_k^{\text{ens}} = \sum_{i=1}^{K} w_i \cdot \hat{y}_k^{(m_i)} \tag{10}$$

where $\hat{y}_k^{(m_i)}$ is the prediction of model m_i for target y_k, and $K = 3$ in this study.

2.6. Model Evaluation Metrics

In this study, various metrics were used to compare the performance of the machine learning models. Through these metrics, the best-performing prediction model was selected, enabling future forecasts to be generated using the most reliable model. The evaluation metrics used in the study can be categorized into two main groups. The first group consists of basic regression metrics, while the second group consists of additional classification-based metrics. The contents of these metric groups are visualized in Figure 7 and explained in detail under the following subsections.

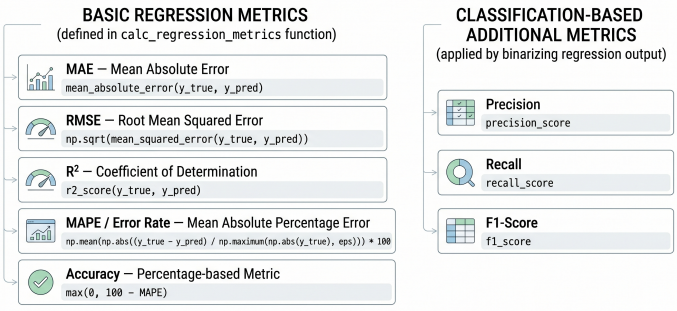


Figure 7: Regression and Classification-Based Metrics Used for Model Evaluation

In the Results section of the article, all metric scores, including Accuracy, MAE, RMSE, R², Error Rate, Precision, Recall, and F1-score, are reported. However, during the visualization stage, only the most important metrics were included in the tables to avoid excessive complexity. The complete source code and detailed results are available for overall evaluation.

2.6.1. Basic Regression Metrics

The performance of all regression models was assessed using five widely adopted error metrics. Following [46], the coefficient of determination (R^2) is considered the most informative among standard regression evaluation criteria, as it provides a normalized, scale-independent measure of explanatory power.

- MAE measures the average magnitude of prediction errors without considering their direction. It is expressed in the same unit as the target variable and is robust to large individual errors [46].

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{11}$$

- Root Mean Square Error (RMSE) penalizes larger errors more severely than MAE due to the squaring process, making it more sensitive to outliers [46].

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{12}$$

- Mean Absolute Percentage Error (MAPE) expresses the prediction error as a percentage of the actual values, enabling scale-independent comparisons across different targets [46].

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{\max(|y_i|, \varepsilon)}\right| \times 100 \tag{13}$$

where ε is a small constant introduced to prevent division by zero.

- Coefficient of Determination (R^2) quantifies the proportion of variance in the target variable explained by the model. A value of 1 indicates a perfect fit, while values close to 0 or negative indicate poor predictive performance [46].

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{14}$$

where $\bar{y}$ denotes the mean of observed values.

- Accuracy is derived from MAPE and represents the percentage of prediction correctness. It is bounded below by zero to avoid negative accuracy values [46].

$$\text{Accuracy} = \max(0,\ 100 - \text{MAPE}) \tag{15}$$

The properties and interpretation of each evaluation metric are summarized in Table 12.

Table 12: Summary of Regression Evaluation Metrics

Metric	Range	Optimal Value	Interpretation
MAE	$[0, +\infty)$	$\to 0$	Average absolute error; same unit as target
RMSE	$[0, +\infty)$	$\to 0$	Penalizes large errors; sensitive to outliers
MAPE	$[0, +\infty)$	$\to 0$	Scale-free percentage error
R^2	$(-\infty, 1]$	$\to 1$	Proportion of variance explained by the model
Accuracy	$[0, 100]$	$\to 100$	Percentage correctness derived from MAPE

2.6.2. Classification-Based Additional Metrics

In addition to regression metrics, classification-based metrics were computed by binarizing the regression outputs to assess the directional accuracy of the forecasts. Specifically, predictions and actual values were converted into binary classes based on whether the value increased or decreased relative to the previous period. Precision, Recall, and F1-score were then applied to evaluate the model's ability to correctly identify the direction of change [47].

- Precision indicates the share of true positive predictions among all samples classified as positive. A higher precision score means that the model makes fewer false positive predictions [47].

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

- Recall represents the fraction of actual positive instances that are correctly recognized by the model. It shows how effectively the model captures true positives while minimizing missed positive cases [47].

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

- F1-Score is the harmonic mean of Precision and Recall, providing a single balanced measure that accounts for both false positives and false negatives [47].

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. The summary of the classification-based evaluation metrics is presented in Table 13.

Table 13: Summary of Classification-Based Evaluation Metrics

Metric	Range	Optimal Value	Interpretation
Precision	$[0, 1]$	$\to 1$	Fraction of correct positive predictions
Recall	$[0, 1]$	$\to 1$	Fraction of actual positives correctly detected
F1-Score	$[0, 1]$	$\to 1$	Harmonic mean of Precision and Recall

3. Results

In this section of the article, the procedures carried out throughout the study are described in detail. The presentation is divided into three main parts, which represent the core stages of the study. The overall workflow of these three parts was presented in Figure 1 in Section 2. The outputs obtained from the model predictions are evaluated in the Discussion section, where the findings are assessed within the overall framework of the study.

3.1. Health Infrastructure Forecast

In the IFM of the study, healthcare-related data for the period between 2025 and 2030 were forecasted using machine learning algorithms based on health data from 2002 to 2024. The module proceeds in five stages: the province-level records are assembled and preprocessed, candidate models are trained and compared on the 2002–2024 period, the best performers for each target are combined into a weighted ensemble, population values for the intermediate forecast years are filled by extrapolation, and the ensemble is finally applied to produce the 2025–2030 projections. The procedures carried out in this part are explained in detail under the following subsections. The flowchart of the IFM is presented in Figure 8.

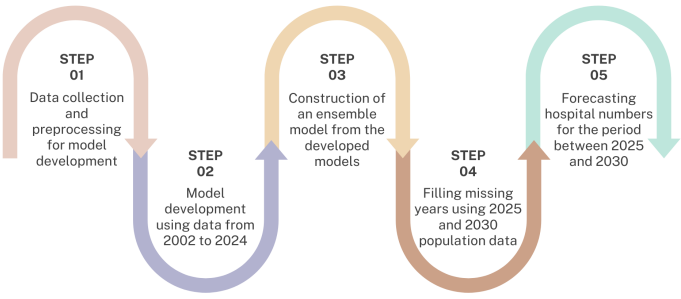


Figure 8: Five-Stage Workflow of the Infrastructure Forecasting Module

3.1.1. Data Preparation

In the first stage of this part, the province-level data on population, number of hospitals, number of hospital beds, and healthcare personnel, as described in the "Data Collection Sources" section, were collected for the period between 2002 and 2024. Since the dataset, comprising 13,041 data points across 1,863 province-year observations, was stored in separate files, these files were merged into a single CSV file as explained in the "Data Preprocessing" section.

3.1.2. Model Development

In the second stage of this part, the data from 2002 to 2024 were used to develop the best-performing prediction model. The purpose of developing these models was to identify the machine learning algorithm that could most accurately predict the available data and, accordingly, generate reliable forecasts for the period between 2025 and 2030.
Before proceeding to the modeling stage, the preparation steps described under the "Pre-Modeling Preparation" section were applied to the data, making the dataset ready for model development. Subsequently, the machine learning methods specified under the "Machine Learning Approaches" section and identified as relevant to the first part of the study were employed. The performance results of the models are presented in Table 14.

Table 14: Comparison of Machine Learning Models for Infrastructure Forecasting, Averaged Across Six Target Variables and Five Walk-Forward Folds

Model	Accuracy (%)	MAE	RMSE	R^2
HoltWinters	**91.594**	**176.692**	**573.802**	**0.972**
ARIMA	90.886	221.379	768.090	0.947
Prophet	90.527	202.769	684.702	0.963
Voting	86.059	247.361	820.054	0.941
ExtraTrees	85.708	240.752	759.110	0.951
CatBoost	85.553	247.939	823.359	0.935
XGBoost	85.447	264.961	895.543	0.922
HGB	85.269	335.039	1426.567	0.829
RF	85.185	263.475	847.860	0.938
GB	85.073	250.023	757.372	0.948
LR	77.584	254.076	526.481	0.940
Ridge	76.197	256.111	526.497	0.940
ElasticNet	75.101	277.180	601.941	0.937

Best-performing values are shown in bold.

3.1.3. Ensemble Model

The results obtained under Section 3.1.2 represent the best overall performance averaged across all target variables. However, this averaging introduces a limitation: a model that predicts the physician variable very accurately but performs poorly on the hospital variable will have a reduced average score and may not be ranked as the best model overall, even though it is highly effective for a specific target. To address this, the three best-performing models were selected separately for each target variable and combined into a weighted ensemble using the approach described in Section 2.5. The resulting ensemble weights and model selections for each target are presented in Table 15.
Following Table 15, the performance metrics of the developed community model are given in Table 16.
It was observed that ensemble modeling produced lower results than individual models for some predictions. However, despite this, it was decided to proceed with the ensemble model. The reason for this decision was the risk of flat-forecast behaviour.

3.1.4. Flat Forecast Behaviour of ARIMA on Sparse Series

When ARIMA encounters a near-constant time series, it collapses to a random walk model and produces flat forecasts equal to the last observed value [48]. In such cases, the model may achieve an artificially high accuracy on historical test data, while failing to generate meaningful forecasts beyond the training horizon [49].
For example, if no new hospital was built in Bolu between 2021 and 2024, and ARIMA correctly predicted the hospital number in 2021, it could continue to predict the same unchanged value until 2024 and therefore achieve an apparently perfect prediction score (Table 17). However, when future forecasts are generated for the 2025–2030 period, ARIMA may continue to produce constant values, which can lead to inaccurate long-term forecasts. For this reason, ensemble modeling was constructed to generate more dynamic and variable predictions.

Table 15: Three Best-Performing Models Selected for Each Target Variable and Their Normalised Ensemble Weights (Fold 5)

Target	Model	Accuracy	MAE	RMSE	Weight
Other	**ARIMA**	**90.598**	**375.373**	**1146.703**	**0.337**
Other	HoltWinters	90.479	379.220	1206.848	0.336
Other	HGB	87.873	607.133	2742.473	0.327
Physician	**HoltWinters**	**91.976**	**231.136**	**685.106**	**0.337**
Physician	ARIMA	90.619	300.441	947.616	0.332
Physician	CatBoost	90.058	283.047	1082.126	0.330
Midwife	**ARIMA**	**93.777**	**34.045**	**68.273**	**0.346**
Midwife	HoltWinters	92.300	39.515	64.822	0.340
Midwife	RF	85.265	82.051	127.816	0.314
Hospital	**ARIMA**	**95.956**	**0.654**	**1.411**	**0.356**
Hospital	HoltWinters	93.197	0.975	1.572	0.346
Hospital	Voting	80.074	2.558	5.838	0.297
Nurse	**ExtraTrees**	**93.937**	**166.521**	**421.617**	**0.335**
Nurse	Voting	93.531	198.314	551.330	0.333
Nurse	XGBoost	93.184	219.763	624.091	0.332
Bed	**ARIMA**	**92.710**	**264.546**	**837.160**	**0.341**
Bed	HoltWinters	92.033	263.947	882.778	0.339
Bed	CatBoost	86.801	273.435	530.161	0.320

Best-performing values are shown in bold.

Table 16: Performance of the Weighted Ensemble by Target Variable

Target	Accuracy	MAE	RMSE	R^2
Other	90.814	354.194	1003.492	0.966
Physician	92.499	258.361	875.994	0.977
Midwife	91.851	42.386	66.102	0.995
Hospital	**92.527**	**0.949**	**1.580**	**0.997**
Nurse	**93.910**	**183.656**	**512.585**	**0.991**
Bed	92.722	207.703	531.969	0.992

Best-performing values are shown in bold.

Table 17: Example of Flat ARIMA Forecasts: Hospital Count in Bolu Province, 2021–2024

Province	Year	Target	Actual	Predicted
Bolu	2021	Hospital	11.0	11.0
Bolu	2022	Hospital	11.0	11.0
Bolu	2023	Hospital	11.0	11.0
Bolu	2024	Hospital	11.0	11.0

3.1.5. Population Projections for 2025–2030

After developing an ensemble model that predicted data for the period 2002–2024 with a mean accuracy of 92%, the next step was to forecast health infrastructure variables for 2025–2030. However, this required province-level population values for each year in the forecast horizon as input features.
As stated in Table 2 under Section 2.1.2, TURKSTAT provides province-level population projections only for the reference years 2025 and 2030. Since intermediate years were not directly available, population values for 2026–2029 were estimated using linear extrapolation based on the historical trend of the last ten years of observed data. For each province p, a linear trend model shown in Equation 19 was fitted using ordinary least squares over the most recent ten observations:

$$\hat{P}_p(t) = a_p \cdot t + b_p \tag{19}$$

where t denotes the year, and the coefficients a_p and b_p are estimated by minimizing the sum of squared residuals over the training window:

$$[a_p, b_p] = \arg\min_{a,b} \sum_{i=1}^{10} \Big(P_{p,i} - (a \cdot t_i + b)\Big)^2 \tag{20}$$

The fitted model was then evaluated at each target year $t \in \{2026, 2027, 2028, 2029\}$ to produce the projected population values. A total of 486 province-year pairs were filled using this extrapolation procedure.

3.1.6. Health Infrastructure Forecast for 2025–2030

Using the previously established weighted ensemble model and the province-level population projections for 2025–2030, health infrastructure variables were forecast at the province level for each year in the target horizon. The forecast was produced for six target variables: number of hospitals, hospital beds, physicians, nurses, midwives, and other healthcare personnel.

The forecasting procedure was applied sequentially for each year $t \in \{2025, 2026, \ldots, 2030\}$. Since each target variable uses the remaining targets as input features, a circular dependency arises during future-year prediction. To resolve this, an iterative update strategy was adopted: for each forecast year, the prediction loop was executed $r = 3$ times, allowing the estimates to converge as each target's predicted value is progressively refined. Formally, at iteration r and year t as given in Equation 21:

$$\hat{y}_k^{(r)}(t) = \sum_{i=1}^{3} w_i \cdot f_i\Big(\hat{y}_{-k}^{(r-1)}(t), P(t)\Big) \tag{21}$$

where $\hat{y}_{-k}^{(r-1)}(t)$ denotes the predicted values of all targets except y_k from the previous iteration, $P(t)$ is the projected population for year t, and f_i is the i-th model in the ensemble. All predictions were clipped at zero and rounded to the nearest integer. The final output consists of $81 \times 6 = 486$ province-year observations covering the 2025–2030 period.

3.2. TOPSIS-Based Provincial Ranking

TOPSIS ranks alternatives according to their geometric closeness to a positive ideal solution (PIS) and separation from a negative ideal solution (NIS) [50]. The best alternative is the one located nearest to the PIS while remaining farthest from the NIS. The method has been widely applied in healthcare facility planning and resource allocation problems [51].

Table 18: The Four Criteria Used in the TOPSIS Ranking and Their Optimisation Direction

Criterion	Unit	Interpretation	Type
0–14 age group population ratio	%	Higher → more demand	Benefit
65+ age group population ratio	%	Higher → more demand	Benefit
Physicians per 1,000 people	per 1,000	Higher → better supply	Cost
Hospital beds per 100,000 people	per 100,000	Higher → better supply	Cost

Supply indicators are treated as cost criteria because higher supply implies lower relative need.

Provincial rankings were constructed using four criteria that reflect both demographic pressure and service capacity, as shown in Table 18. Criterion weights were determined objectively using the Shannon entropy method, which identifies the most discriminating criterion based on the distribution of values across provinces [50]. The TOPSIS procedure was applied as follows:

1. Data collection and preprocessing
2. Entropy-based criterion weighting
3. Model development using the available data
4. Future forecasting using the developed model
5. Sequential Resource Allocation with Feedback

3.2.1. Data Collection and Preprocessing

During the data collection stage, the 0–14 and 65+ population data for the period between 2007 and 2025 were obtained from TURKSTAT, as described in Section 2.1. In addition, the number of physicians per 1,000 people and the number of hospital beds per 100,000 people for the period between 2002 and 2024 were collected. Province-level population data for the period between 2002 and 2024 were also included. Following the data collection process, the preprocessing steps described in Section 2.2 were applied, and the files were combined into a single CSV file. The number of data records used for the TOPSIS model is presented in Table 19.

Table 19: Number of Data Records Used for the TOPSIS Model

Variable	Period	Records
0–14 age group population ratio	2007–2025	1,539
65+ age group population ratio	2007–2025	1,539
Physicians per 1,000 people	2002–2024	1,863
Hospital beds per 100,000 people	2002–2024	1,863
Province-level total population	2002–2024	1,863
Total		**8,667**

3.2.2. Entropy-Based Criterion Weighting

Shannon entropy, originally introduced in the context of information theory, quantifies the uncertainty or information content within a probability distribution [52, 53]. A criterion with low entropy exhibits high variability across alternatives, meaning it carries more discriminating power. Conversely, a criterion with high entropy shows little variation and contributes less to the ranking [54]. This property makes Shannon entropy a widely used objective weighting method in multi-criteria decision-making, as it derives weights directly from the data without requiring subjective judgment.

In this study, criterion weights were calculated from the 2025 provincial data using the following procedure. Let x_{ij} denote the value of criterion j for province i, and let m be the number of provinces ($m = 81$).

Step 1 — Normalization (Equation 22):

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^{m} x_{ij}} \tag{22}$$

Step 2 — Entropy calculation:

$$E_j = -k \sum_{i=1}^{m} p_{ij} \ln(p_{ij}), \quad k = \frac{1}{\ln(m)} \tag{23}$$

Step 3 — Divergence:

$$d_j = 1 - E_j \tag{24}$$

Step 4 — Weight assignment:

$$w_j = \frac{d_j}{\sum_{j=1}^{n} d_j} \tag{25}$$

The resulting weights are presented in Table 20. The 65+ age ratio received the highest weight (37.0%), reflecting the greatest inter-provincial variability and thus the strongest discriminating power among the four criteria.

Table 20: Entropy-Based Criterion Weights

Criterion	Weight	(%)
65+ age group population ratio	0.3698	37.0
0–14 age group population ratio	0.1976	19.8
Physicians per 1,000 people	0.2205	22.1
Hospital beds per 100,000 people	0.2121	21.2
Total	**1.0000**	**100.0**

3.2.3. Model Development Using the Available Data

The same ML models described in Section 2.4 were applied to the four TOPSIS criteria using a five-fold walk-forward cross-validation scheme. The primary objective was to identify the best-performing model for each criterion and subsequently generate forecasts through 2030. Each target variable was trained on its own available historical range: 2007–2025 for age group ratios and 2002–2024 for the clinical supply indicators.

Initial results were strong for two criteria, with Ridge selected for the 65+ age ratio and LR for physicians per 1,000. The 0–14 age ratio and the hospital bed ratio, however, yielded low or negative R^2 values across all models. For these two targets the feature set was therefore revised: lagged values of the target were added and any variable used to construct the target was excluded to prevent leakage. Baseline_Lag1, a naive predictor carrying forward the previous year's value, was also added to the candidate pool as a reference. Because of this revision, the two targets were evaluated on the three most recent years (2022–2024) rather than the five-fold walk-forward scheme used for the other criteria. HGB then performed best for the 0–14 age ratio and ElasticNet for the hospital bed ratio, both exceeding the Baseline_Lag1 reference. Table 21 presents the final model selected for each criterion.

Table 21: Best-Performing Model Selected for Each TOPSIS Criterion (Fold 5 for the first two criteria; 2022–2024 evaluation for the last two)

Target	Model	Acc. (%)	MAE	RMSE	R^2
65+ age ratio	Ridge	99.28	0.072	0.085	0.864
Physicians per 1,000	LR	98.94	0.026	0.027	0.954
0–14 age ratio	HGB	99.39	0.128	0.191	0.999
Beds per 100,000	ElasticNet	98.70	3.540	4.280	0.997

3.2.4. Future Forecasting Using the Developed Models

Using the best-performing model identified for each criterion, provincial forecasts were generated for the 2025–2030 period. Ridge was used for the 65+ age ratio, LR for physicians per 1,000, HGB for the 0–14 age ratio, and ElasticNet for hospital beds per 100,000.

3.2.5. Sequential Resource Allocation with Feedback

A critical issue arises when TOPSIS scores are computed statically across the entire 2025–2030 horizon: a province with persistently high need indicators would receive the largest budget share every year, regardless of whether prior allocations have already begun to address its deficiencies. To overcome this, a dynamic feedback mechanism was adopted. After each annual allocation, the TOPSIS criteria values are updated to reflect resources already directed to each province, and the ranking is recalculated for the following year. This ensures that provinces whose needs have been partially met yield priority to others with greater remaining deficiencies, as illustrated in Figure 9.

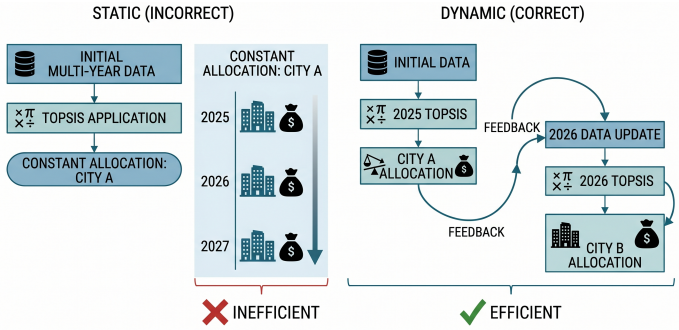


Figure 9: Static Versus Dynamic TOPSIS Allocation: Fixed Ranking Compared with Annual Recalculation

3.3. Health Budget Modeling

The third component of this study focuses on forecasting Turkey's national health budget for the 2025–2030 period. The forecasts were produced in USD to avoid distortions caused by Turkish lira depreciation, and subsequently converted to TRY using projected exchange rates. The subsections below detail the data collection, exchange rate construction, model development, and future budget forecasting steps. The workflow of the Health Budget Forecast section is shown in Figure 10.

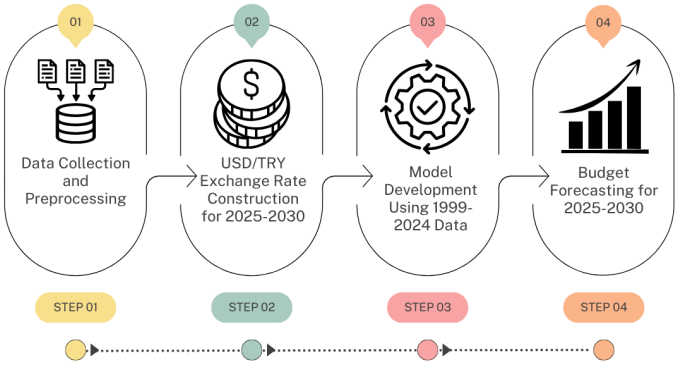


Figure 10: Four-Stage Workflow of the Health Budget Forecasting Module

3.3.1. Data Collection and Preprocessing

In this section, the data collection and preprocessing steps for budget forecasting are described. As outlined in Section 2.1, the dataset consists of Turkey's annual general health expenditure figures for the period 1999–2024 and the monthly USD-TRY exchange rate for the same period. Following data collection, the preprocessing steps described in Section 2.2 were applied. The number of records used is presented in Table 22.

Table 22: Number of Data Records Used for Budget Modeling

Variable	Period	Records
Turkey annual health expenditure	1999–2024	25
Monthly USD/TRY exchange rate	1999–2024	300
Total		**325**

3.3.2. USD/TRY Exchange Rate Construction for 2025–2030

Since forecasting Turkey's health budget directly in Turkish lira would introduce distortions due to currency depreciation, all budget figures were converted to USD, requiring the construction of a monthly USD/TRY exchange rate series for the 2025–2030 period. Data collection process is described in Section 2.1, and Table 23 summarizes the number of actual versus estimated monthly observations by year.

Table 23: Monthly USD/TRY Data Composition by Year

Year	Actual (months)	Forecast (months)	Source
2025	12	0	X-Rates, ExchangeRates.org
2026	5	7	Mixed
2027	0	12	LongForecast + interpolation
2028	0	12	LongForecast + interpolation
2029	0	12	LongForecast + interpolation
2030	0	12	LongForecast + interpolation
Total	**17**	**55**	

The resulting annual average USD/TRY rates used in budget conversion are presented in Table 24.

Table 24: Annual Average USD/TRY Exchange Rates (2025–2030)

Year	USD/TRY
2025	39.54
2026	46.32
2027	55.93
2028	66.08
2029	83.17
2030	96.18

3.3.3. Model Development Using 1999–2024 Data

The health expenditure figures were extracted from the 1999–2024 dataset, comprising annual total health expenditure, per capita expenditure, and the public and private sector expenditure ratios. Since the Turkish lira has undergone significant depreciation over this period, forecasting directly in TRY would inflate apparent growth rates. Therefore, all budget figures were converted to USD using the corresponding annual average exchange rate. Several time-series features were derived to preserve the temporal structure of the data, including the previous year's budget, the two-year lagged budget, a rolling moving average, year-on-year growth rates, and a linear trend index. Eight models were trained on this dataset, covering time-series methods (ARIMA, Holt, HoltDamped), compound annual growth rate models (CAGR5, CAGR10), and tree-based regressors (RF, ExtraTrees, GB). Model performance was evaluated using a within-sample test scheme, with particular emphasis on forecast accuracy in the most recent years to assess forward-looking reliability, as reported in Table 25.

Rather than relying on a single best-performing model, the top five models were combined into a weighted ensemble to reduce the risk of overfitting to any individual model's strengths and to improve forecast stability. Ensemble weights were assigned proportionally to each model's Fold-5 accuracy, so that higher-performing models contributed more to the final prediction. The resulting ensemble included a diverse mix of tree-based regressors, a time-series model, and growth rate estimators, providing complementary perspectives on the budget trajectory. The selected models and their ensemble weights are shown in Table 26.

Table 25: Comparison of Models for National Health Budget Forecasting (USD Target)

Model	Accuracy (%)	MAE	RMSE
ExtraTrees	**90.210**	**4,965**	**6,157**
CAGR10	88.370	5,571	6,826
GB	88.254	5,662	6,884
CAGR5	88.153	5,633	6,962
ARIMA	88.114	5,559	6,618
HoltDamped	86.677	6,011	7,148
RF	86.786	6,127	7,349
Holt	86.119	6,256	7,407

Best-performing values are shown in bold.

Table 26: Models Selected for the Budget Forecasting Ensemble and Their Weights

Model	Accuracy (%)	Ensemble Weight
ExtraTrees	**90.210**	**0.2201**
CAGR10	88.370	0.1962
GB	88.254	0.1930
CAGR5	88.153	0.1940
ARIMA	88.114	0.1966

Best-performing values are shown in bold.

3.3.4. Budget Forecasting for 2025–2030

The weighted ensemble model constructed in the previous section was applied to generate Turkey's national health budget forecasts for the 2025–2030 period. Since forecasting directly in Turkish lira would introduce distortions due to currency depreciation, all projections were produced in USD. The ensemble combined the outputs of ExtraTrees, ARIMA, CAGR10, CAGR5, and GB models, with weights proportional to each model's Fold-5 accuracy. The resulting USD figures were subsequently converted to TRY using the projected annual average exchange rates presented in Table 24.

3.4. Integration of TOPSIS Rankings and Budget Allocation

In the final stage, the 2025–2030 TOPSIS provincial priority rankings were combined with the forecasted national health budget. Rather than applying a static allocation based on a fixed ranking, a dynamic feedback mechanism was adopted. After each annual budget distribution, the received funds are treated as a partial improvement in a province's healthcare capacity, thereby reducing its need score in subsequent years. To quantify this effect, a budget factor with a multiplier of 0.25 was introduced, meaning that prior allocations account for 25% of the updated TOPSIS score in each following year. The value of 0.25 was selected to reflect a moderate adjustment rate, ensuring that prior allocations meaningfully reduce a province's relative need score without dominating the demographic and supply indicators that form the core of the TOPSIS ranking. This corresponds to a weighting scheme in which received resources account for one quarter of the updated score, preserving the primacy of structural need over historical funding patterns. This ensures that provinces which have already received substantial funding yield priority to others with greater remaining deficiencies. The complete allocation cycle is illustrated in Figure 11.

3.5. Evaluation of Results

The dynamic TOPSIS-based allocation yielded different ranks for the provinces for the period 2025–2030, which indicates that the feedback mechanism is capable of avoiding a fixed set of provinces to dominate the budgetary resources year after year.

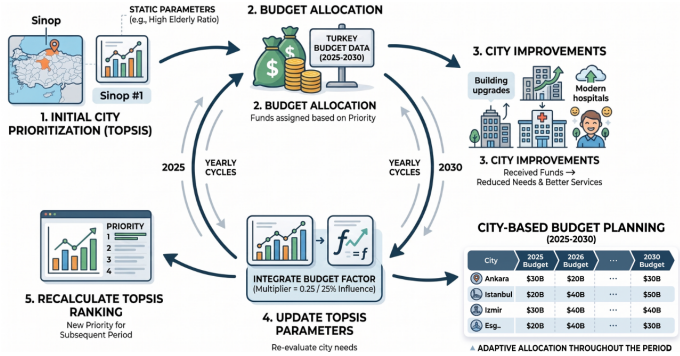


Figure 11: Annual Cycle of the Dynamic Budget Allocation Mechanism

In 2025, the prioritized provinces were mostly small cities in the north of Anatolia such as Sinop, Artvin, Kastamonu, Çankırı and Bartın. The common profile of these provinces is that they have high ratios of 65 years old and above and low physician density, which is why they record high TOPSIS scores. Their small populations also resulted in very high per capita allocations (Artvin and Sinop approximately 282,800 TRY and 214,000 TRY per person respectively in 2025).

These large per-capita transfers had the direct effect of providing the top-ranked provinces with enough resources in 2025 to reduce their relative need scores. From 2026 on, the ranking changed significantly and Balıkesir rose to the first position, where it stayed until 2030, with Aydın, Muğla, Manisa and Mersin in the upper ranks. These are larger coastal provinces with aging populations and moderate clinical supply indicators that were previously displaced by smaller, more severely underserved cities. Their larger populations result in lower per capita allocations.

Table 27 summarises the top five provinces by TOPSIS priority rank for each year of the forecast horizon. The dynamic feedback mechanism causes a notable structural shift after the first forecast year: while small northern provinces dominate in 2025 due to high elderly population ratios and low supply coverage, accumulated budget support from 2026 onward redistributes relative need, bringing larger western coastal provinces to the top of the priority list. Provinces receiving large early investments thus progressively yield their top positions to others with greater remaining deficiencies, which aligns with the core objective of equitable and need-responsive resource distribution.

Table 27: Top 5 Priority Provinces by Year, Dynamic TOPSIS (2025–2030)

Year	Province	TOPSIS Score	Per Capita (TRY)
2025	**Sinop**	**0.712**	**214,167**
	Artvin	0.697	282,800
	Kastamonu	0.689	123,239
	Çankırı	0.689	233,376
	Bartın	0.661	217,190
2026	**Balıkesir**	**0.826**	**40,089**
	Aydın	0.810	42,911
	Ordu	0.806	65,728
	Muğla	0.799	44,663
	Manisa	0.797	33,646
2027	**Balıkesir**	**0.794**	**51,987**
	Aydın	0.774	55,248
	Ordu	0.763	84,487
	Muğla	0.761	56,739
	Manisa	0.760	43,397
2028	**Balıkesir**	**0.782**	**64,016**
	Aydın	0.762	67,848
	Muğla	0.750	69,017
	Manisa	0.748	53,507
	Ordu	0.747	104,047
2029	**Balıkesir**	**0.780**	**83,548**
	Aydın	0.759	88,248
	Muğla	0.748	88,925
	Manisa	0.745	69,917
	Mersin	0.742	51,187
2030	**Balıkesir**	**0.781**	**100,262**
	Aydın	0.758	105,496
	Muğla	0.749	105,401
	Manisa	0.745	84,049
	Mersin	0.744	61,288

Within each year, provinces are listed in descending order of TOPSIS score, so the first row corresponds to the province with the greatest assessed need. The highest-ranked province of each year is shown in bold.

4. Discussion

4.1. Comparison with the Literature

The forecasting results of this study diverge from those reported in comparable work. Previous studies on healthcare workforce and expenditure forecasting in Turkey identified tree-based or kernel-based learners as the strongest performers: XGBoost for physician supply and demand [6], Random Forest for nurse staffing [7] and for per capita health expenditure [16], and Support Vector Regression for national health spending [17]. Here, time-series methods outperformed these families across all six infrastructure targets, with HoltWinters, ARIMA, and Prophet reaching 91.59%, 90.89%, and 90.53% accuracy respectively, against 85.71% for the best tree-based model (Table 14). This divergence follows from the structure of the data. Each province-level series contains only 23 annual observations and exhibits a strong monotonic trend driven by sustained capacity expansion since 2002. Tree-based learners predict piecewise-constant values within observed ranges and cannot extrapolate beyond the training range, a limitation visible in the lowest coefficient of determination in the comparison ($R^2 = 0.829$ for HGB). The cited studies operate on cross-sectional data or on national aggregates over longer horizons, conditions under which this constraint does not bind, so the present findings complement rather than contradict them. Time-series methods carry their own failure mode on near-constant series, which the weighted ensemble was adopted to contain (Section 3.1.4).

A second body of work applies entropy-weighted TOPSIS to health system evaluation [9–14]. These studies establish that entropy-based objective weighting suits health-sector indicators, and the present study adopts the same weighting principle. They differ from it in temporal orientation: each ranks units on observed data at a single point in time, and the resulting ranking does not respond to any allocation that follows from it, as is also the case for the provincial ranking in [8]. The consequence has been documented directly, since spatial inequalities in hospital bed distribution across 2308 Chinese counties persist where allocation rules do not adjust to prior investment [15]. The feedback mechanism proposed here addresses this condition by reducing a province's need score in proportion to funds already received. Table 28 quantifies its effect.

Table 28: Effect of the Feedback Mechanism on the Circulation of Provincial Priority (2025–2030)

Indicator	Static	Dynamic
Distinct provinces entering the top five	5	11
Years before the top five first changes	—	1
Provinces ranked first at any point	1	2

Under a static ranking computed once on 2025 data, the same five provinces would hold the top five positions in every year of the horizon. Under the dynamic scheme, eleven distinct provinces enter the top five and the composition changes structurally after the first allocation year, as the small northern provinces prioritised in 2025 yield to larger western coastal provinces whose deficiencies remain unaddressed.

Persistence is reduced rather than eliminated: Balıkesir remains first from 2026 to 2030, reflecting a demographic and supply profile that a single feedback cycle does not resolve.

4.2. Practical Implications

The entropy weighting assigned 37.0% of the total criterion weight to the 65+ population ratio, against 19.8% for the 0–14 ratio and a combined 43.3% for the two supply indicators (Table 20). Because entropy weights derive from inter-provincial variability rather than expert judgement, this indicates that the elderly population share is currently the strongest discriminator among Turkish provinces. For planning purposes it implies that allocation criteria weighted primarily by total population or by absolute personnel deficits will systematically understate need in provinces undergoing demographic ageing, a pattern concentrated in the northern and central Anatolian provinces that dominate the 2025 ranking. The per capita figures reported in Table 27 express relative priority rather than operationally feasible transfers, and in practice would require bounding by absorptive capacity; the framework accommodates such constraints without structural modification, since the allocation stage is separable from the ranking stage.

4.3. Limitations and Future Work

The proposed framework yields promising results in the context of province level healthcare resource planning, although some limitations need to be taken into account. First, the exchange rate projections used to express the budget forecasts in Turkish lira were obtained from third party forecasting sources for the 2026–2030 period. Since long-term currency forecasts involve considerable uncertainty, these projections may affect the accuracy of TRY-based allocation results, especially for later years in the forecast horizon. Second, the feedback multiplier (0.25) for the dynamic TOPSIS allocation process was chosen by expert judgment rather than empirical optimization. Different values of the multiplier could affect the rate of change of priority classes across provinces, so testing the sensitivity of this parameter is another important avenue for future research.

Another limitation is that the proposed framework has not been compared to real budget allocation decisions made by the Ministry of Health. Therefore, further evaluation of its practical performance under real planning conditions is still needed.

5. Conclusion

This study developed an integrated ML framework for province-level healthcare resource planning in Turkey for the 2025–2030 period, addressing three interrelated problems: infrastructure forecasting, need-based provincial ranking, and national budget projection. Walk-forward cross-validation across 19 models showed that time-series methods, particularly HoltWinters, ARIMA, and Prophet, consistently achieved the highest accuracy for infrastructure targets, while a weighted ensemble strategy was adopted to mitigate the flat-forecast behaviour of ARIMA on sparse series. TOPSIS-based rankings identified small northern provinces such as Sinop, Artvin, and Kastamonu as facing the greatest healthcare need in 2025, driven by high elderly population ratios and low physician density, with Shannon entropy assigning the strongest discriminating power to the 65+ age ratio. The dynamic feedback mechanism prevented budget monopolisation, shifting priority to larger underserved provinces such as Balıkesir, Aydın, and Muğla from 2026 onward as early recipients reduced their relative need scores. Budget forecasts were produced in USD to avoid distortions from Turkish lira depreciation, using an ensemble of ExtraTrees, ARIMA, CAGR, and GB models that achieved 88–90% accuracy.

The study makes three contributions. First, it provides a province-level infrastructure forecast covering all 81 Turkish provinces across six indicators, generated under a validation scheme that preserves temporal order. Second, it introduces an entropy-weighted TOPSIS ranking that is recalculated annually in response to allocations already made, so that priority circulates rather than concentrating in a fixed set of provinces. Third, it links a national budget forecast to province-level allocations, connecting macroeconomic projection to sub-national distribution within a single framework.

Several directions remain open. The feedback multiplier was set by expert judgement and would benefit from validation-based optimisation. Long-term exchange rate uncertainty could be represented through stochastic scenarios rather than point projections. Comparing the framework's allocations against actual Ministry of Health decisions would establish its practical value, and the same three-module structure could be applied to other countries with comparable provincial health statistics. The framework is designed to be re-executed annually, so projections can be replaced by realised values as they become available.

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